

Does Inter-Stimulus Distance Influence the Decoding Performance of SSVEP and SSMVEP BCI?

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Abstract—Brain-computer interfaces (BCIs) based on steady-state visually evoked potentials (SSVEP) are considered as one of the most successful noninvasive BCI paradigms. Recently, an evolution of SSVEP was introduced with repetitive motion as visual stimuli instead of the conventional flickering stimuli. This novel paradigm is steady-state motion visually evoked potential (SSMVEP), and it has been proposed to address some inherent limitations of the flickering stimulus. One important factor that affects the decoding performance of BCIs based on repetitive visual stimulus is the change in inter-stimulus distance (ISD) or competing stimuli. In this study, a preliminary analysis was performed to compare the influence of change in ISD on the decoding performance of SSVEP and SSMVEP paradigms. The experiments were performed on thirteen healthy participants in which two visual stimulus types: flickering stimuli and checkerboards stimuli, were presented. Three different ISDs were studied. The decoding performance was quantified by the canonical correlation analysis (CCA) on O1, O2, and Oz channels. With 1-second window length, the average decoding accuracy across all ISDs was 75% for the SSVEP stimulus and 83.6% for the SSMVEP stimulus, respectively, and the corresponding average variation across ISDs was 0.13 and 0.03. The results demonstrated striking robustness of SSMVEP against competing stimuli while maintaining an overall superior accuracy to that of the conventional SSVEP.

I. INTRODUCTION

A Brain-computer interface (BCI) enables direct communication or control between the brain and peripheral devices by acquiring neural signals corresponding to various brain activities, analyzing collected signals, and translating them into information corresponding to the intentions of the user [1]. The most commonly used type of BCIs are noninvasive and are based on electroencephalogram (EEG), which measures the electrical activity of cortical neurons from the surface of the scalp. BCIs based on steady-state visually evoked potentials (SSVEP) have been extensively studied and considered as one of the most successful noninvasive BCI paradigms. SSVEPs are EEG components induced by repetitive visual stimuli, and are prominently observed in the occipital and parieto-occipital region of the cortex [2]. When the visual stimulus's luminous intensity is modulated at a certain frequency, the EEG's dominant frequency from the user's occipital area will be the same as

the stimulus frequency. When multiple visual stimuli with different target frequencies are placed in the user's visual field, the visual stimulus with the user's engagement can be detected by detecting the most dominant spectral peak in the EEG. For example, SSVEP-based BCI was used to decode covert intentions of paralyzed patients and enable them to communicate with and control external devices [3]. SSVEP-based BCIs are known for their high decoding accuracy, and consequently, high information transfer rate (ITR) [4], [5]. Another advantage of SSVEP BCIs is that they can be used without any training or calibration, comparing with many other BCIs that require lengthy subject training [6], [7]. The low subject-training of SSVEP BCIs makes them highly appealing for many real-world applications [8].

One of the key limiting factors of SSVEP BCIs is that the flickering stimuli can easily cause visual fatigue and other effects, *e.g.*, headaches, nausea, and even increased risk of seizures [9], and researchers have been addressing some of these challenges [10]. Additionally, an extension of the SSVEP-like paradigm using repetitive motion called steady-state motion visually evoked potential (SSMVEP) has been recently proposed [11]. The SSMVEP BCI has been reported to cause lower visual fatigue and mental load than SSVEP and follows a different neural pathway in the visual cortex from SSVEP to output information [12], [13].

Another inherent limitation of any SSVEP-like stimuli is the effect of competing stimuli. When multiple visual stimuli are presented simultaneously within the same visual field, the stimuli other than the target stimulus compete for neural representations, which subsequently causes interference in the signal and influences the decoding performance [14]. Apparently, the closer the stimuli are, the greater the effect from the neighboring competing stimuli [15] [16].

In this study, a preliminary analysis was performed to compare the influence of change in inter-stimulus distance (ISD) on the decoding performance of SSVEP or flickering stimulus and SSMVEP or circular oscillating checkerboards. To modulate the effect of competing stimuli, three different ISDs were studied. The performance was measured based on the classification accuracy in a seven-class scenario. The rest of the paper is organized as follows: Section II presents the methodologies, including the stimulus design and experiments, the EEG analysis and BCI performance analysis. Section III presents the results from the experiment and discusses the results and Section IV concludes the paper.

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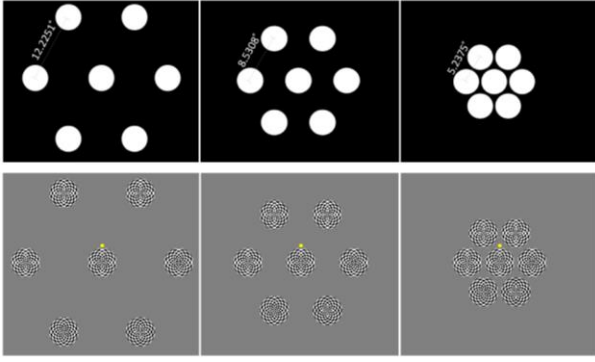


Figure 1. Stimulus Design D₃, D₂, and D₁ for SSVEP (flicker) stimulus (top) and SSMVEP (checkerboard) stimulus (bottom) respectively.

II. METHODOLOGY

A. Stimulus Design

In this study, two types of visual stimulus-based BCI paradigms were compared: the classical flicker stimulus for SSVEP and the circular oscillating checkerboard stimulus for SSMVEP. For consistency, the stimulus parameters such as stimulus shape, size, ISD, and stimulating frequency were kept the same for both stimulus types. Seven visual stimuli were displayed on an LED monitor screen (with refresh rate of 60 Hz), the stimulus frequencies were carefully synchronized with the refresh rate of the monitor. For both visual stimulus types, the seven frequencies were: 8.423 Hz, 9.375 Hz, 9.961 Hz, 10.84 Hz, 11.87 Hz, 13.4 Hz, and 14.87 Hz, respectively. The stimulus orientation was such that one stimulus was placed at the center of the screen, and six others were placed concentrically around the central stimulus.

The LED monitor was placed in front of the participant at a distance of 0.6 m, measured from the participant's eyes. To investigate the effect of changing ISD on BCIs' performance, three ISDs were implemented for each stimulus type. The ISD was measured as the visual angles between any individual stimuli were: 5.2375°, 8.5308°, and 12.2251°, similar to those implemented in our previous study [16].

Moreover, the stimulus parameters specific to each type of stimulus were designed as discussed further. The flickering or SSVEP stimuli were circular with a five cm diameter and were alternating between black and white as in [16]. The circular oscillating checkerboard, *i.e.* SSMVEP stimuli, were also five cm in diameter and was designed similar to the study [17]. Each circular oscillating checkerboard was divided into 24 sectors with 10 rings. Finally, all stimuli were designed and implemented using the PsychoPy toolkit, an open-source Python package designed for psychology experiments [18]. An example of the two stimulus types: SSVEP or flickering stimulus and SSMVEP or checkerboard stimulus for the three different ISDs used in this study are shown in Figure 1.

B. Data Acquisition and Experimental Protocol

Thirteen subjects participated in this study, where participants self-reported having normal or corrected-to-normal vision. The study was approved by the Office of Research Ethics of the University of Waterloo (ORE #31850). Each participant signed a written informed consent before

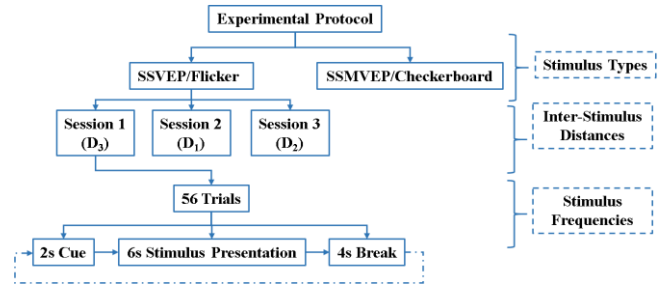


Figure 2. Experimental protocol illustrating the procedure for one stimulus type and one ISD.

starting the experiment. After signal quality analysis, data from twelve participants were kept for further analysis, and one data set was removed from the analysis due to poor signal quality.

The g.USBamp and Gammabox (g.tec Guger Technologies, Austria) wet electrode (g.Scarabeo) system (sampling rate of 1,200 Hz) was used to collect the EEG signals, and OpenVibe software was used to acquire the data. The electrodes were placed in the following six locations: O_1 , O_2 , O_z , PO_3 , PO_4 , PO_z according to the International 10-20 system and the EEG was acquired. The right earlobe was used to place the reference electrode, and a ground electrode was placed on FPz. In this study, only three channels, O_1 , O_2 and O_z were used for the analysis.

Figure 2 illustrates the experimental protocol used in this study. Two types of visual stimulus at three different ISDs were compared in this study. Therefore, three sessions for each of the two visual stimulus types were conducted. One of the three ISDs was randomly presented for each session. Within each session, a total of 56 stimulus presentations were displayed each of 12 s duration with 2 s cuing with a yellow marker, 6 s stimulus displaying flickering or radial oscillation, and 4 s rest period. The order of the runs for the different stimulus types, sessions, and trials were all randomized and counter-balanced during the experiment, and between two consecutive sessions a 3-minute break was provided. The data was recorded and processed for offline analysis.

C. Canonical Correlation Analysis (CCA)

The baseline method Canonical Correlation Analysis (CCA) was used to quantify the BCI performance level change with changing ISDs. CCA is the most commonly used baseline algorithm for SSVEP/SSMVEP target detection. CCA is an unsupervised method used to measure the correlation between two sets of data [19]. All signals were preprocessed by applying a 4th order Butterworth band-pass filter between 1Hz – 40Hz. In this study, the SSVEP/SSMVEP trials were processed asynchronously: each trial was segmented based on a window obtained between [0.5s, 2s] from the onset of the stimulus with an incremental shift of 0.1s, and CCA was applied to each segment.

CCA was used to calculate the correlation between a segment of the stimulus epoch and a reference signal containing sinusoids of frequencies corresponding to the first and second harmonics of each target stimulus frequency. As

a result, the target stimulus frequency is inferred by identifying the highest correlation coefficient and was identified as the target with the user's visual attention. The next section defines the standard CCA procedure.

Consider the multi-channel EEG signal during a stimulus presentation epoch as: $X = (x_1, x_2, \dots, x_n)$, where n is the number of EEG channels. The set of reference vectors for the i^{th} stimulus frequency would be:

$$Y_i = \begin{bmatrix} \cos(2\pi \cdot f_i \cdot t) \\ \sin(2\pi \cdot f_i \cdot t) \\ \dots \\ \cos(2\pi \cdot h \cdot f_i \cdot t) \\ \sin(2\pi \cdot h \cdot f_i \cdot t) \end{bmatrix}, t = \left[\frac{1}{f_s}, \frac{2}{f_s}, \dots, \frac{m}{f_s} \right], \quad (1)$$

where f_i is the target stimulation frequency, f_s is the sampling rate, m is the number of samples and h is the number of harmonics where $h = 2$ in this study. Given the EEG signal and reference signals, the objective of CCA is to solve (2) by finding the weight vectors w_x and w_y . This is performed by maximizing the correlation $\text{corr}(w_x^T X, w_y^T Y)$. The correlation coefficients ρ are calculated as:

$$\rho_i = \max_{w_x, w_y} \frac{w_x^T \text{cov}(X, Y) w_y}{\sqrt{w_x^T \text{cov}(X, X) w_x} \sqrt{w_y^T \text{cov}(Y, Y) w_y}} \quad (2)$$

D. BCI Performance Analysis

The decoding performance was measured based on the CCA classification. CCA applied on the target segment results in a vector of correlation coefficients. Therefore, the highest coefficient, *i.e.* ρ_i , will be selected from the coefficient vector ρ , and the i^{th} frequency, consequently the target segment will be assigned the class label of the i^{th} stimulus. Finally, all decoding results for all trials were used to calculate the accuracy. The accuracy was calculated for each visual stimulus type, ISD, and window size. Additionally, analysis of variance (ANOVA) was performed on the computed results to compare the influence of change in ISD on the decoding performance of the SSVEP and SSMVEP BCIs. The stimulus type with two levels, window length with four levels, and ISDs with three levels were the fixed factors, and the decoding accuracy was the response variable. The null hypothesis was that the decoding accuracies were the same for all three ISDs. The significance level was set as 0.05.

III. RESULTS AND DISCUSSIONS

The decoding accuracy for both stimulus types under three different ISDs was calculated across all window lengths and all subjects. These results are summarized in Figure 3. The figure illustrates the decoding accuracy for the SSMVEP on the left panel and for the SSVEP on the right panel across all ISDs and window sizes. The results indicated that across all window lengths between [0.5 s, 2.0 s] and across the three ISDs, the overall decoding accuracy was higher for the SSMVEP stimulus than the SSVEP stimulus.

The statistical analysis revealed that all main effects such as stimulus type, ISD and window length had a significant (p

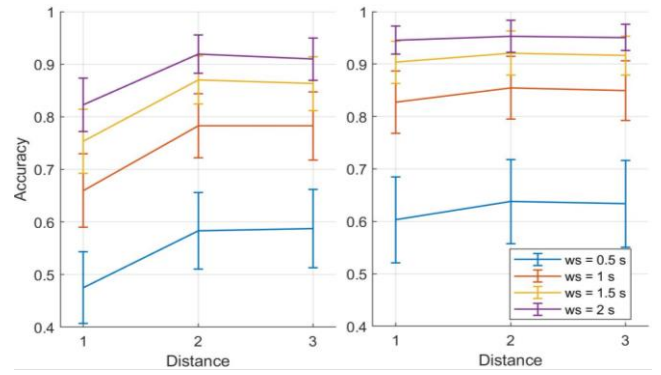


Figure 3. Comparing the decoding accuracy of SSVEP (f-flicker) stimulus (left) and SSMVEP (c-checkerboard) stimulus (right) for analysis window lengths (ws) between [0.5s, 2s] across the three distances $D_1 < D_2 < D_3$.

$= 10^{-3}$) influence on the decoding accuracy. The interactions between the stimulus type and window length ($p = 0.0394$), as well as stimulus type and ISD were both statistically significant ($p = 10^{-3}$). No other 3-way or 2-way interactions were found to be significant. The statistical analysis showed that the difference in the performance between D_2 and D_3 was not significant for both SSVEP ($p = 0.8271$) and SSMVEP ($p = 0.7193$). For D_1 and D_3 , it was significant for SSVEP ($p < 10^{-3}$) and not significant for SSMVEP ($p = 0.1091$). Similarly, between D_1 and D_2 , it was significant for SSVEP ($p < 10^{-3}$) and only marginally significant for SSMVEP ($p = 0.0501$).

A drastic increase in performance can be observed when increasing the distance from D_1 to D_2 for the SSVEP stimuli, whereas for the SSMVEP stimuli there is stable and higher decoding accuracy. Figure 4 illustrates the average variation in decoding performance computed as the difference between ISDs D_1 and D_3 . It can be observed that the average variation in the decoding performance is significantly lower for the SSMVEP compared to SSVEP. The variation is higher for SSVEP of more than 0.11 for window length < 2 s, while for SSMVEP it is less than 0.04 for all window lengths, and decreases further with larger windows. This result suggests that the SSVEP BCI is more sensitive to change of ISD compared to the SSMVEP BCI.

The performance of an SSVEP/SSMVEP BCI system can be influenced by many factors including stimulus design, detection algorithm, analysis windows, etc. In this study, most stimulus parameters were fixed, and the influence of change in ISD at different analysis windows for two types of visual stimuli on the decoding performance of the BCI was studied.

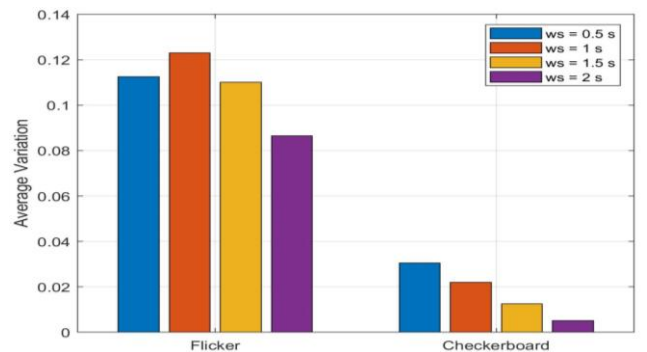


Figure 4. Comparing the average variation across ISDs (difference between D_1 - D_3) of four analysis window lengths (ws) between [0.5s, 2s].

Therefore, with the change in ISD or the presence of competing stimuli, a preferable visual stimulus paradigm should produce higher decoding accuracy and have less impact due to the competing stimuli. The analysis presented in this study showed that the SSMVEP outperformed SSVEP under all window sizes and ISDs. The results provide evidence to show that the SSMVEP stimulus is a good visual stimulus paradigm that is simultaneously robust to change in ISD and also provides high decoding accuracy.

Additionally, prior studies have reported that the SSVEP or flicker stimulus is more likely to induce visual fatigue and mental load than SSMVEP or checkerboard stimulus under the same stimuli luminance contrast. This could explain the higher performance and less variation due to competing stimuli in the SSMVEP or checkerboard stimulus type [12]. As expected, the closest ISD with significant competing stimuli had the worst performance in both stimulus types. However, an unexpected but interesting observation was that the performance of D₂ ISD was always similar to or even slightly higher than the D₃ ISD in most of the scenarios investigated, which is different from our previous experiment [16]. The previous study used EEG segments with 3 s window length, starting from 0.5 s from the beginning of each stimulation, while in this study we investigated shorter periods. This observation could be attributed to higher cognitive attention and eye movements required for the D₃ ISD resulting in comparatively inferior BCI performance. Future studies can investigate this aspect further, some other features.

IV. CONCLUSIONS

In this study, some preliminary results comparing the effect of competing stimuli or inter-stimulus distance on the decoding performance of SSVEP and SSMVEP BCI was presented. This study investigated the following question: does change in inter-stimulus distance influence the decoding performance of SSVEP and SSMVEP BCI? Our results demonstrated that answer is affirmative. Further, such effect was significantly higher for SSVEP BCI than for SSMVEP BCI. These results indicate that SSMVEP BCI is more robust against changing levels of competing stimuli and its performance is less affected by the change of ISD. Future work will investigate and provide more detailed analysis of the EEG spectrum of the resultant trials across both stimulus types and ISDs.

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